Arista Access Node IND-ACC-65 Indiranagar Exchange Optical Transport Failure Procedure

Node: IND-ACC-65 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_7191

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Arista Access Node IND-ACC-65 at the Indiranagar Exchange site is a critical issue that requires immediate attention. This failure domain occurs due to the degradation of the SFP/XFP transceivers or fiber cuts, leading to a loss of optical signal strength and, subsequently, a failure of the link. The underlying protocol mechanics involve the use of Optical Transport Network (OTN) protocols, such as ODUk and OPUk, which are responsible for transmitting optical signals over the fiber links. When the SFP/XFP transceivers or fiber links degrade, the OTN protocols fail to function correctly, resulting in a loss of link connectivity. This failure domain is specific to the Access role and Arista equipment due to the vendor-specific implementation of OTN protocols and the use of SFP/XFP transceivers. The severity of this failure is classified as Warning (ALARM_7191) due to the potential impact on network connectivity and service availability.

Pre-requisite Checks

- Pre-requisite: Verify the site location and node ID to confirm the affected node is IND-ACC-65 at the Indiranagar Exchange site.

- Pre-requisite: Check the redundant link status to ensure that the secondary link is operational and can take over in case of a failure.

- Pre-requisite: Ensure that remote hands are dispatched to the Indiranagar Exchange site to assist with physical access to the node.

Potential Root Causes

Cause	Probability
SFP/XFP Transceiver Degradation	High
Fiber Cut or Damage	Medium
Optical Signal Strength Loss due to Environmental Factors	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface tengigabitethernet 1/1-2, Expected Output: Tengigabitethernet1/1: notconnect, HW, 10W1, 1000BaseKR SFP, Full, Description: This command is used to verify the operational status of the tengigabitethernet interface, which is responsible for transmitting the optical signal. The expected output indicates that the interface is not connected, which could be due to the SFP/XFP transceiver degradation or fiber cut.
```

```
Command: show optics tengigabitethernet 1/1-2, Expected Output:
Tengigabitethernet1/1: 1000BaseKR SFP, Rx Power: -20.5 dBm, Tx Power:
-3.2 dBm, Description: This command is used to verify the optical
signal strength and power levels of the tengigabitethernet interface.
The expected output indicates that the Rx power is lower than the
normal range, which could be due to the SFP/XFP transceiver
degradation or fiber cut.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP Transceiver, Command: replace tengigabitethernet 1/1-2 with tengigabitethernet 1/1-2, Description: This step involves replacing the SFP/XFP transceiver with a new one to resolve the issue. The command is used to replace the transceiver on the tengigabitethernet interface.

Step 2: Step: Verify the Optical Signal Strength, Command: show optics tengigabitethernet 1/1-2, Description: This step involves verifying the optical signal strength and power levels of the tengigabitethernet interface after replacing the SFP/XFP transceiver. The command is used to verify the optical signal strength and power levels.

Post-Check Verification

```
Command: show interface tengigabitethernet 1/1-2, Expected Output:
Tengigabitethernet1/1: connected, HW, 10W1, 1000BaseKR SFP, Full,
Description: This command is used to verify the operational status of
the tengigabitethernet interface after resolving the issue. The
expected output indicates that the interface is connected, which
confirms that the issue is resolved.

Command: show optics tengigabitethernet 1/1-2, Expected Output:
Tengigabitethernet1/1: 1000BaseKR SFP, Rx Power: -15.5 dBm, Tx Power:
-2.5 dBm, Description: This command is used to verify the optical
signal strength and power levels of the tengigabitethernet interface
after resolving the issue. The expected output indicates that the Rx
power is within the normal range, which confirms that the issue is
resolved.
```

Roll-back Procedures

- Step: Restore the Original SFP/XFP Transceiver, Command: replace tengigabitethernet 1/1-2 with tengigabitethernet 1/1-2, Description: This step involves restoring the original SFP/XFP transceiver to resolve the issue in case of a failure. The command is used to restore the original transceiver on the tengigabitethernet interface.